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In 2005, scientists from Bhabha Atomic Research Centre unveiled a revolutionary design for a Thorium Breeder Reactor, which can produce 600 MW of electricity for two years, with no refuelling and practically no control manoeuvres

From the labs

alternative fuels for a different type of nuclear reactor, but as this fuel does not lead to the creation of weapon making material as a waste product, the governments are not interested in using it. The name of this fuel is Thorium, which is a radioactive material, named after Thor, the Norse god of thunder.

The promise of Thorium

Between 1964 and 1969 United States had done major experiments for using Thorium for producing energy. The results were positive, but after the initial tests were completed the Thorium reactors were shut down and the focus returned solely on Uranium based reactors. This was the cold war era and USA needed sufficient amount of nuclear weapons, which could only be had from a Uranium fuel cycle. Once again the attention is returning to Thorium. India is in the forefront of the countries, which are currently engaged in developing new technologies that can use thorium fuel cycle to produce electricity. According to some estimates, about 25 per cent of the world's known reserves of thorium are located in India. Only Australia has more known reserves of the element. In fact, Thorium is available in earth's crust in such large quantities that it poses as a virtually inexhaustible source of energy. Almost all the Thorium that is mined can be used as a fuel, whereas in case of Uranium, the ratio stands at around 1 per cent.

Thorium occurs naturally as Th-232 and it has a half life of over 14 billion years, or about three times the age of earth. The best thing about thorium

fuel cycles is that they produce much less radioactive waste, including the weapon grade plutonium, as compared to uranium and hence are easier on the environment. The fact that it is impossible for a terrorist group or a rogue state to use Thorium reactors to make nuclear weapons does have the effect of making this technology extremely appealing to peace loving citizens. After all, the idea of having nuclear power without the inherent risks of radioactive pollution and threats of proliferation has some kind of a political appeal all over the world. During the last two decades, scientists from India and few other countries have been playing a major role in developing Thorium-based reactors. The Indian government is expecting that by 2050, almost 25 per cent of India's energy needs will come from nuclear sources, a part of which will be fuelled by thorium.

Carlo Rubbia, who won the Nobel Prize in physics in 1984, has been working on developing the use of Thorium as a cheap, clean and safe fuel. Recently Carlo Rubbia had said, "A tonne of the silvery metal could be used to produce as much energy as 200 tonnes of Uranium, or 3,500,000 tonnes of coal. A mere fistful would light London for a week." According to scientists, the biggest challenge in building a thorium reactor comes from its legendary efficiency. Thorium is so efficient as a fuel that its reactions can be sustained for a very long time. So the fuel needs to be enclosed in special containers that are durable enough to withstand the corrosive salts. Numerous ideas for how such special containers could be

built by using corrosion-resistant alloys and graphite have been put forward, but such systems are yet to be tested.

Cartels delaying the launch of Thorium

More than anything else, it is establishment prejudice that seems to be hindering the development of thorium fuelled reactors. As the entire world's nuclear infrastructure is built around uranium, many large corporates could loose out on their huge investments, if thorium based reactors became popular. The international trade in uranium reactors and in enriched fuel runs into hundreds of billions of dollars; thorium could prove to be a game changer and pave way for the rise of a new kind of reactors, as well a fuel that is much cheaper and is easily available. Purely financial motives could be motivating powerful international uranium cartels to use their clout for hindering research into thorium. Despite the initiatives taken by India and few other countries, there still exists the risk that thorium may not be taken up by the broader nuclear energy industry. That would be unfortunate. It makes lot of sense for governments and private organisations to seriously pursue thorium as a source of unending green energy.

Kirk Sorensen, a former NASA rocket engineer and now chief nuclear technologist at Teledyne Brown Engineering, considers Thorium to be the next big thing. He says, "Once you start looking more closely, it blows your mind away. You can run civilisation on thorium for hundreds of thousands of years, and it's essentially free. You don't have to deal with uranium cartels." Some scientists have suggested that one way of tackling the lack of funds could be to start using thorium in uranium based reactors. But Kirk Sorensen is against the idea. He says, "You have to use the right machine. You don't use diesel in a petrol car: you build a diesel engine." Once unique plants for using thorium as fuel become operational, nuclear power could become much more routine and unthreatening. Thorium could be the elusive material that will one day allow emerging economies like India and China to provide energy to their massive populations over their creaking infrastructures. 

